ETIOLOGY AND PATHOGENESIS

Although several genes have been identified as the cause of inherited forms of vascular malformations, the etiology of the most common sporadic lesions is unknown. Histologically, vascular malformations consist of enlarged, tortuous vessels with quiescent endothelium. In contrast to hemangioma, there is no parenchymal mass nor cellular proliferation.

CLASSIFICATION

For many years, vascular anomalies were grouped under the term *angioma*, which hampered precise classification and led to incorrect diagnosis and improper management. For example, the term *hemangioma* has been used both for vascular malformations—often of the venous type (cavernous hemangioma)—and for vascular tumors (strawberry hemangioma). This changed in 1982 with the development of a biologic classification by Mulliken and Glowacki, which divided vascular anomalies into two major categories: **vascular tumors** (characterized by cellular proliferation; hemangioma being the most common) and **vascular malformations** (structural anomalies of blood vessels). Vascular malformations are further subdivided based on the predominant vessel type into arterial, capillary, lymphatic, or venous malformations (VMs). In 1996, this classification was adopted and refined by the International Society for the Study of Vascular Anomalies.

Vascular malformations most commonly affect a single vessel type, but combined forms exist and are named according to the involved vessels (e.g., capillary-venous or veno-lymphatic malformation). In addition to isolated cases, vascular malformations can occur as part of syndromes such as:

- **Klippel-Trenaunay syndrome**: capillary-lymphatico-venous malformation with limb hypertrophy
- **Maffucci syndrome**: multiple enchondromas associated with multiple venous malformations and an increased risk of malignancy
- **Parkes Weber syndrome**: high-flow vascular malformation of the extremity with soft tissue hypertrophy

VASCULAR MALFORMATIONS

AT A GLANCE

- **Worldwide prevalence**: ~0.3% of the population
- **Congenital**, though may present later in life
- **Localized, well-demarcated lesions** composed of malformed vessels (capillary, venous, lymphatic, arteriovenous)
- **Histology**: Enlarged, tortuous vessels without endothelial proliferation
- May be **isolated**, **combined**, or part of a **syndrome**
- **Management**: Requires a multidisciplinary approach

CLINICAL FINDINGS

Vascular malformations grow proportionally with the patient and typically do not regress. Most are well-demarcated and localized. Rarely, they may represent cutaneous markers of deeper underlying lesions.

They are classified hemodynamically into:

- **Slow-flow malformations**: capillary, lymphatic, venous, and combined types

- **Fast-flow malformations**: arteriovenous and arterial malformations (with or without venous involvement)

Doppler ultrasound is the preferred non-invasive imaging modality for initial evaluation, providing information on flow characteristics, vessel architecture, and extent of the lesion.